

1. Use `ggplot2` to construct a scatter plot of the relationship between distance traveled and average speed in miles per hour of every flight that left the bay area in 2020 (recall the data set is called `flights`). Give your plot a title. Write the code you used below, and also, make a rough sketch of your visualization.
2. Is the problem of predicting average speed by distance traveled a regression problem or a classification problem? How about using distance to predict whether a plane travels over 400 miles per hour?
3. Use the `lm()` function to fit a linear model by least squares that predicts the average speed of the flight based on the distance and save it to the object `m1`. Write your code below.
4. Write out the equation of the linear model that you have fit.

- STAT 20 SUMMER 2025